

A Comprehensive Study on Technological Advancements in India's Financial Sector

RITHIKA S, Dr. SANTOSH NELAMAKANAHALLI CHIKKAMARI

Affiliated with Shakti Research Centre and Academia (SRCAA) Contributing Scholars

Category: Banking & Financial Technology | Published: March 2026 | Peer Review: Double-Blind Validated

ABSTRACT

Technology has played a critical role in the development of the Indian banking industry, which has undergone significant changes over time. The study examines the evolution and impact of technology in India's banking industry, focusing on digital advancements that have revolutionized the sector. It analyzes the adoption of technological solutions like mobile banking, internet banking, digital payments, and blockchain technology, and their transformation of traditional banking practices. The study also addresses challenges and opportunities in technology integration, such as cybersecurity, data privacy, regulatory compliance, and the digital divide. It also highlights the role of government, regulators, and industry stakeholders in fostering a conducive environment for technological innovation and ensuring a level playing field for all players. The study also provides insights into future prospects and disruptions that emerging technologies like artificial intelligence, machine learning, and fintech startups may bring to India's financial sector, including increased automation, personalized services, and new business models. Overall, the study offers a comprehensive analysis of how technology

Keywords: Indian Banking Industry; Digital Payments; Mobile Banking; Fintech Disruptions; Cybersecurity & Regulation

1. Introduction & Contextual Background

The transformation of India's banking and financial landscape over the past two decades represents one of the most dynamic technological shifts in emerging economies. From core banking automation to UPI and Account Aggregators, technology has democratized financial access.

2. Technology Adoption & Infrastructure Architecture

Key infrastructural pillars including the India Stack, open API architectures, and cloud-native banking platforms have enabled exponential transaction scalability exceeding 130 billion annual operations.

3. Regulatory Frameworks, Cybersecurity & Governance

As digitalization accelerates, Reserve Bank of India (RBI) mandates around data localization, tokenization, and zero-trust security ensure financial system integrity against rising digital threats.

4. Conclusion & Strategic Recommendations

Technological advancement in India's financial sector will continue to be driven by artificial intelligence and smart contracts. Regulators and financial institutions must collaborate to safeguard consumer privacy.

References & Scholarly Citations

1. RITHIKA S (2026). A Comprehensive Study on Technological Advancements in India's Financial Sector. SRCAA Global Review of Contemporary Research (SGRCR), 1(1), pp. 43-56.
2. Sharma, R., & Gupta, M. (2025). Contemporary Trends in Multidisciplinary Research & Innovation. Academic Press, New Delhi.
3. UNESCO (2023). Global Standards for Open-Access Scientific Publishing and Open Science Frameworks. Paris: UNESCO.
4. Committee on Publication Ethics (COPE). (2024). Core Practices for Academic Journal Transparency and Digital Archiving.

EDITORIAL ARCHIVING & OPEN ACCESS COMPLIANCE STATEMENT

Published by Shakti Research Centre and Academia (SRCAA) under the Creative Commons Attribution 4.0 International (CC BY 4.0) License. Readers are permitted to copy, redistribute, remix, and build upon this work with appropriate scholarly attribution. This document is officially catalogued in the SGRCR website permanent server repository for ISSN validation, DOI indexing, and peer-reviewed scholarly preservation.